Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu



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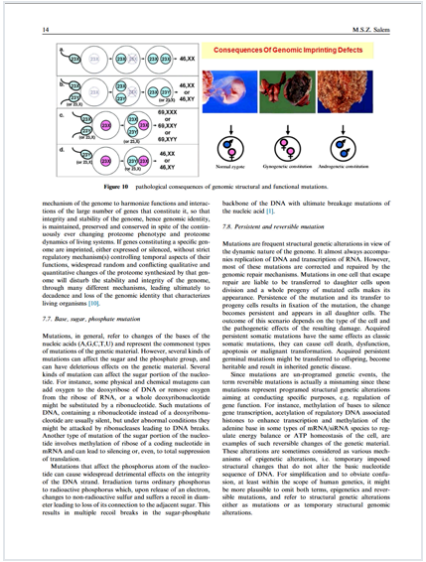
Table of Contents

- **Introduction**
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method

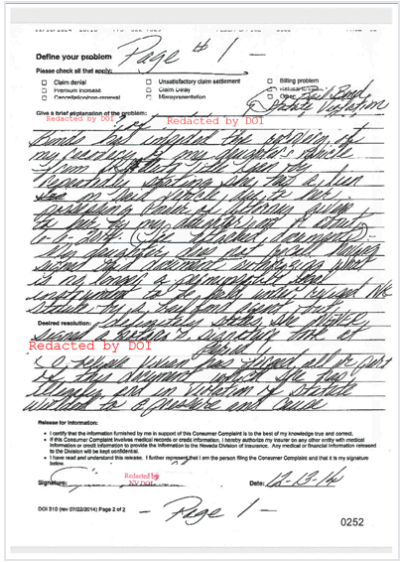
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Multi-Page Visually Rich Documents

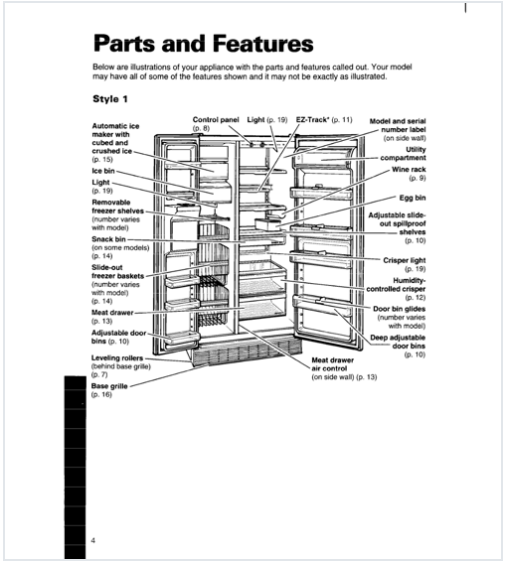
Documents carry meaning through **text, visual information and layout** together. **Multi-page** documents add another layer: meaning can run across pages, as when a sentence refers to a figure several pages away.



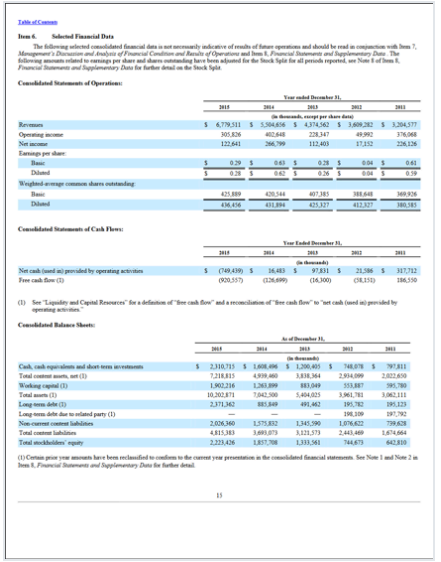
Research paper
born-digital



Signed form
scanned, handwritten



Technical manual
layout-dependent



Financial report
dense tables



Product brochure
photo-led

- **Text.** Typeset or handwritten, embedded in the file or only visible in a scan.
- **Visual information.** Charts, figures, photographs and handwriting.
- **Layout.** How information is arranged in the two-dimensional space of a page.

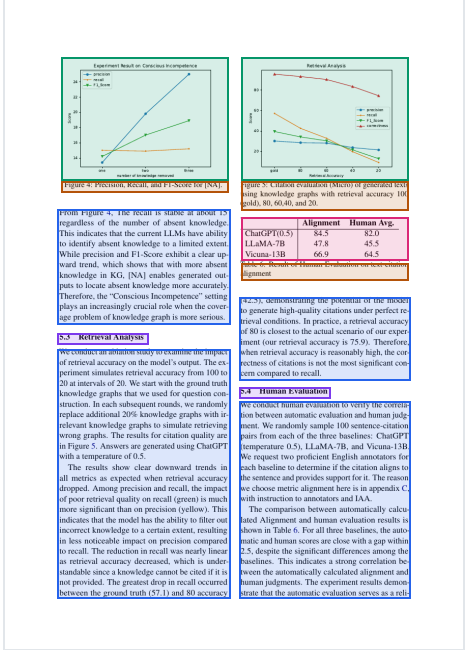
What is Document Understanding?

Answering natural-language questions: **identify the relevant content** and **connect the evidence**, within or across pages.

Long documents make this **very difficult**: evidence is sparse, spread across many pages, and often more than a model can read at once.

LAYOUT ANALYSIS





DOCUMENT OUTLINE

Research paper, 24 pages

- 5.2 Conscious Incompetence
  - Paragraphs
  - Figure 4 — Caption
- 5.3 Retrieval Analysis
  - Paragraphs
  - Figure 5 — Caption
- 5.4 Human Evaluation
  - Paragraphs
  - Table 6 — Caption

EXAMPLE QUESTION

"Does the model rated best by humans in Table 6 also have the best citation recall in Table 3?"

Evidence needed:

- Table 6, this page
- Table 3, previous page

Cross-page: it compares two tables on different pages, so neither alone is enough.

Title · Section heading · Text · Figure / chart · Table · Caption / note · Header / footer

Table of Contents

- Introduction
- **Survey of MP-VRDU**

  Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

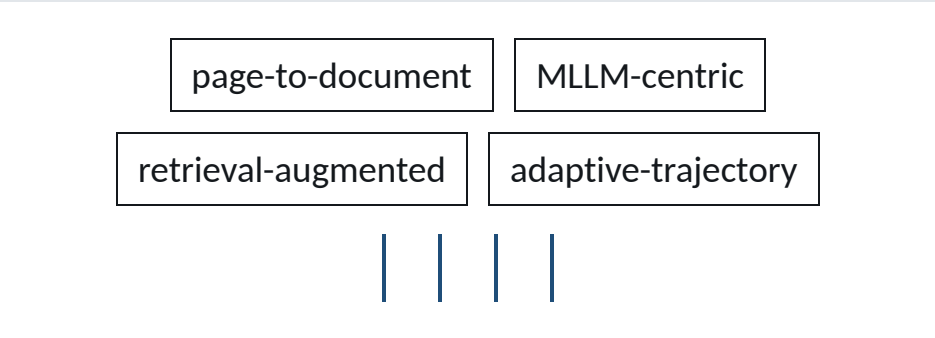
  Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

  EMNLP 2026, main conference
- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method

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Survey: Research Gap and Contributions

RESEARCH GAP



Multi-page systems developed along **parallel lines**, with different terminology and little comparison between them.

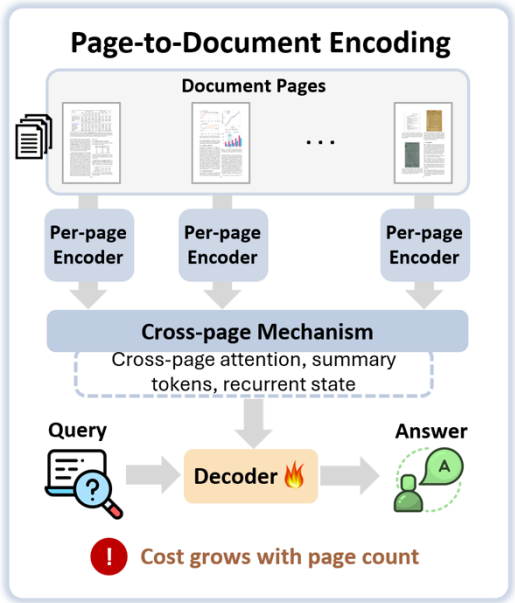
CONTRIBUTIONS

1. Brings these systems together through **evidence management**: how a system manages the relevant evidence across a very long document.

2. Organises them into **four architectural families**.

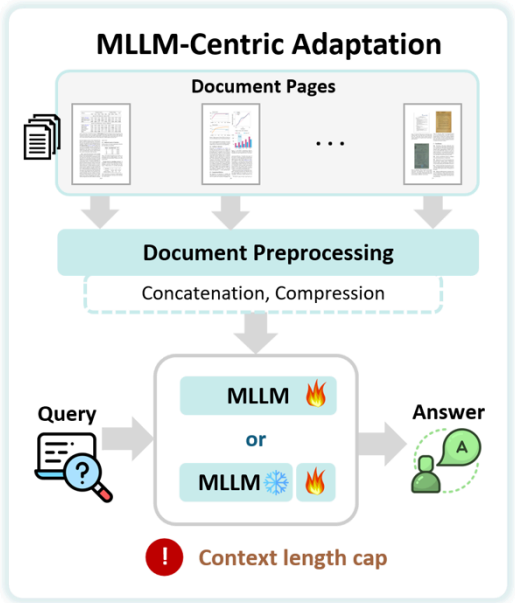
Multi-Page Architectural Overview

Four architectural families. They emerged roughly in this order, although their development overlaps.



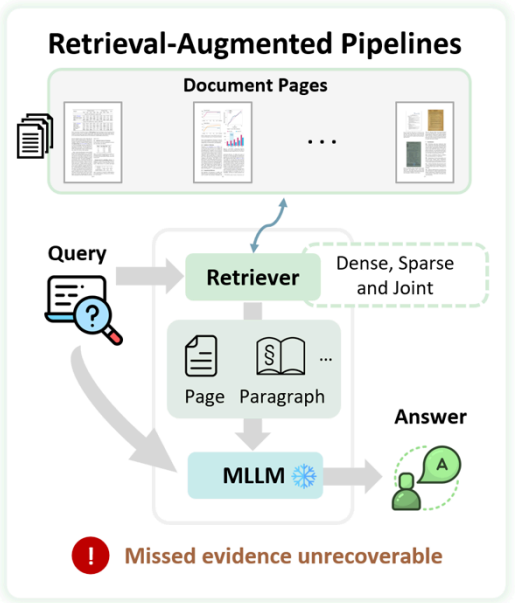
Late 2023 to mid 2024

Encode each page, connect them with a cross-page mechanism, train end to end



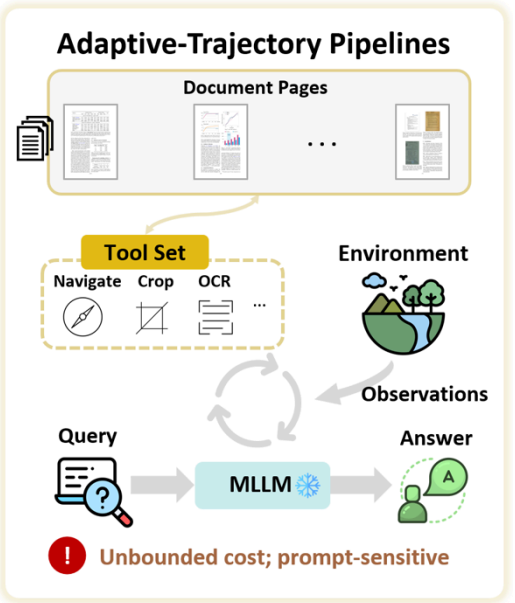
Early 2024 to mid 2026

Feed all pages into a pretrained model's context, with further training



Early 2024 to mid 2026

Retrieve evidence first, then answer. The workflow is fixed in advance



Early 2025 to mid 2026

A language model decides what to inspect next from what it has found

Snowflake indicates typically frozen, fire indicates typically trained

From Research to Products: LLM Assistants

How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?

Read a file, updated tasks

Added task Locate pages for Figure 4.2 and Table 4.2

Request
Description
Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually

Request
Description Read PDF pages 60-61 and extract the bar heights and token statistics

Added task Verify and answer

Request
Description Cross-check figure values against surrounding text, then answer

Completed task Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

Read Honours___Lewei_Xu.pdf

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

Completed task Verify and answer

- Research in this field is applied **directly in public-facing products**.

UNTIL EARLIER THIS YEAR

MLLM-centric adaptation

The full document text and the page images are fed to the model.

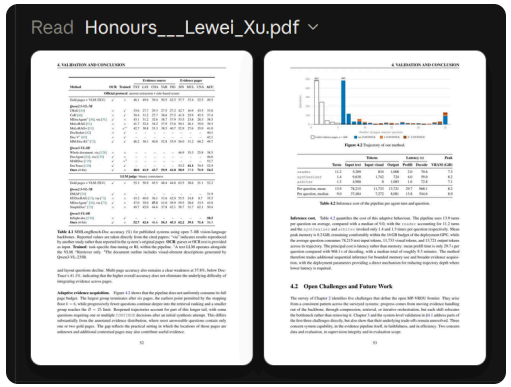
→

MORE RECENTLY

Retrieval-augmented or adaptive-trajectory

The assistant plans, locates the relevant pages, and inspects them before answering.

- Here: a difficult question over my thesis, almost **100 pages**. It **renders the pages as images** and reads them visually.



Surveyed Systems at a Glance

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VT5 (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2025)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B-Instruct	V	-	CoT, HD	-
Leopard (2024)	TMLR	No	SigLIP-SO400M	LLaMa-3.1-8B, Mistral-7B	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B / Qwen2.5-VL-7B	V	-	HD	-
Docopilot (2025b)	CVPR	No*	InternViT-300M	Multiple (Intern)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2-7B-Instruct	V	-	-	-
DocSLM (2025)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025b)	CVPR	Yes	-	LLaMA3-8B, Qwen1.5-7B	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2025)	MMAsia	No	Pix2Struct	wen2.5-VL-3B-AWQ	V	D	-	-
DFVC (2026)	Electronics	No	VisRAG, ColQwen	Qwen2-VL-2B-Instruct	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Multiple (Qwen/Intern/Idefics)	V	D	-	-
MM-R5 (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025b)	EMNLP	No	ColPali	Multiple (Qwen/DS/LLaVA)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	preprint	No	ColPali, Qwen3-VL ViT	Multiple (Qwen/GLM/Gemini)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Multiple (Intern/Pali/Phi)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	MM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024c)	AAAI	Yes	-	Multiple	T, L	S, IR, Nav	-	1A
MHier-RAG (2025)	preprint	Yes	Qwen-VL-Plus	Multiple (Qwen/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Multiple (Qwen)	T, V, L	D	CoT, HD	-
MultiDocFusion (2025)	ACL	Yes	-	Multiple (Qwen/LLaMa/Mistral)	T, V, L	D	HD	-
M2RAG (2025a)	NeurIPS	Yes*	VisRAG-Ret	Multiple (Qwen)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Multiple (Llama/Qwen)	T, V	J	-	5A
PDF-WuKong (2024)	preprint	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B-Instruct	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Multiple(Qwen/Gemini/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	preprint	No	Qwen2.5-VL	Qwen2.5-VL-7B-Instruct	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025a)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Multiple (Qwen/MiMo/Intern)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2026)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B-Instruct	V	D, IR, QR	ReAct, HD	1A
DocLens (2025a)	preprint	Yes	–	Multiple (GPT/Gemini/Claude)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	GPT-4o / Qwen-plus	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	–	Multiple (GPT/Gemini/Claude)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B-A3B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	preprint	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Multiple (Qwen)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025a)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Multiple (Qwen/LLaMa/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

1 ***Models and pipelines***

The first two families are **models**: trained end to end, with components built for that architecture. The last two are **pipelines**: components are often interchangeable, **especially the backbone**.

2 ***Modalities***

Encoding models combine **text, vision and layout**. Models in the second family mostly use **vision only**.

3 ***Search***

Only appears once **retrieval is separated from answering**. Retrieval-augmented pipelines add a retrieval step.

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- Survey of MP-VRDU
- **Empirical Attribution Study**

  Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

  Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

  Under review
- Conclusion
- Proof of Concept: A Simple Method

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Real-World Constraints

In industry: one specific task, over **thousands to millions of documents**.

Privacy

Privacy and data-residency requirements.

No third-party API.

Cost

Even without privacy concerns, API costs add up at this volume.

Scale

Either way, you cannot upload documents **one by one** into an online tool.

API or local model, you have to **build your own document understanding pipeline** that orchestrates a vision-language model, **under a compute budget**.



So where do such pipelines fail? We formalise failure into **three failure modes**.

Empirical Study: Research Gaps and Contributions

RESEARCH GAPS

1. **Competing design claims**, each tested on a different pipeline and dataset.

   vision only **vs** text + vision

   retrieve more **vs** retrieve less

2. **No end-to-end empirical study** of where document understanding systems fail.

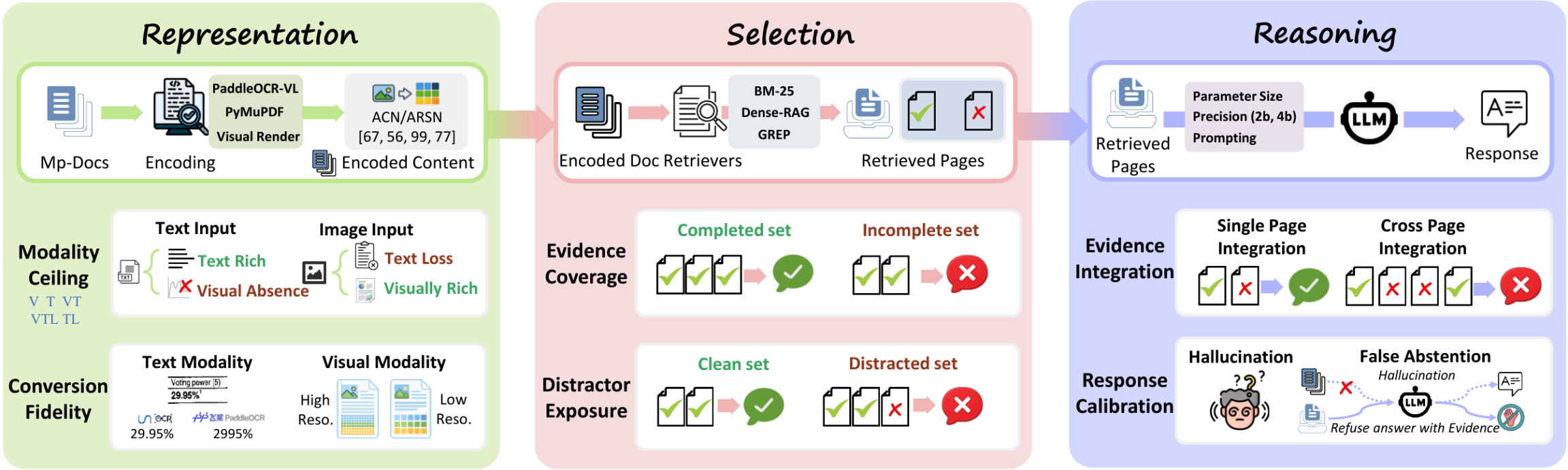
CONTRIBUTIONS

1. A **unified view of failure**: three architecture-independent loci, representation, selection and reasoning.

2. Controlled interventions in **one pipeline**, across document collections and independently developed model families.

3. **Practical guidance** for designing and deploying systems **under a compute budget**, from qualitative examples and accuracy-cost comparisons.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

One simple pipeline. **Change one stage, hold the other two fixed.**

Encode
REPRESENTATION

Each page becomes text, an image, or both.

default: parser text + page image

→

Retrieve
SELECTION

The top pages for the question reach the model.

or: the annotated gold pages

→

Answer
REASONING

One model answers in a single pass.

default: Qwen3-VL-8B

- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Representation: What Should the Model Read?

ASSUMPTION ~~Give the model the richest input, or perfect one modality.~~

FINDING **Text and vision cover for each other, so neither has to be perfect.**

Accuracy spread across parsers

text alone



15.7 pts

with page image



4.2 pts

The image recovers what the parser lost.

Same accuracy, fewer input tokens

parser text + image



1,333

embedded text + image



959

49.5% against 50.1%, and no parser needed.

Time to prepare the input

text only



1.8 s

with images



26.3 s

Per question. Images are not free.

EXAMPLE

Q What are the colors of the icons that the users touch to move the app up and remove the app?

Gold gray, red

Text only	✗	"The document does not specify the colors"
With image	✓	"gray (a gray upward arrow) … red"

DESIGNING AND DEPLOYING UNDER A BUDGET

Start from embedded text
It is cheap, and often reliable on born-digital documents.

▶

Add the page image
When the task needs appearance, or the text looks unreliable.

▶

Re-parse only if needed
When the existing text is inadequate, such as on scans.

Richer input costs time and memory on every page, so **choose the modality and the number of pages together**.

Selection: Which Pages Should the Model See?

ASSUMPTION ~~Retrieving more pages is always safer.~~

FINDING **Missing evidence is costly, but extra pages hurt too.**

Remove one required page

all required pages



38.6%

one missing



18.4%

Retrieval rank does not show which page is safe to drop.

Add three unneeded pages

required pages only



43.9%

plus three more



35.1%

Multi-hop questions, with every required page still present.

Recall at five pages

BM25, on CPU



58.1%

ColQwen3, on GPU



81.0%

0.3 ms against 110 ms per query.

EXAMPLE

Q How many chapter authors of this report come from Columbia University?

Gold 3

Page 13	✗	“two”
Page 14	✗	“one”
Both pages	✓	“three”

DESIGNING AND DEPLOYING UNDER A BUDGET

Retrieve broadly
Coverage first: a missed page cannot be recovered later.

▶

Filter or rerank
Control what actually reaches the model.

▶

Judge the set
Is the whole set sufficient? If not, retrieve again.

A strong visual retriever pays off on collections queried often. A light text retriever suits tight budgets. **Judge by final answers and latency, not recall alone.**

Reasoning: Can the Model Use What It Sees?

ASSUMPTION ~~A larger model fixes reasoning.~~

FINDING **Scale lifts everything, but the multi-hop gap stays, and refusal is a trade-off.**

Gap between single-hop and multi-hop

at 2B parameters

22.0 pts

at 32B parameters

24.5 pts

Both improve by over 20 points. The gap does not close.

Refusing answerable questions

no instruction

6.2%

may abstain

27.7%

reason, then abstain

19.9%

Helps one model family far more than the other.

Storing the 8B model at 4-bit

memory, 16-bit

17.5 GB

memory, 4-bit

6.8 GB

Accuracy 52.4% to 51.8%, but slightly slower.

EXAMPLE

Q How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

Gold 16.7%

May abstain ✗ "Not answerable"

Reason first ✓ "79.9% − 63.2% = 16.7%"

DESIGNING AND DEPLOYING UNDER A BUDGET

Choose the backbone on your task
Model family matters as much as size.

▶

Quantise to fit memory
A larger model at lower precision, if time allows.

▶

Spend reasoning selectively
Extra steps and verification only for complex or uncertain cases.

Calibrate refusal on the model you deploy, and report false refusals alongside accuracy.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

Thank you

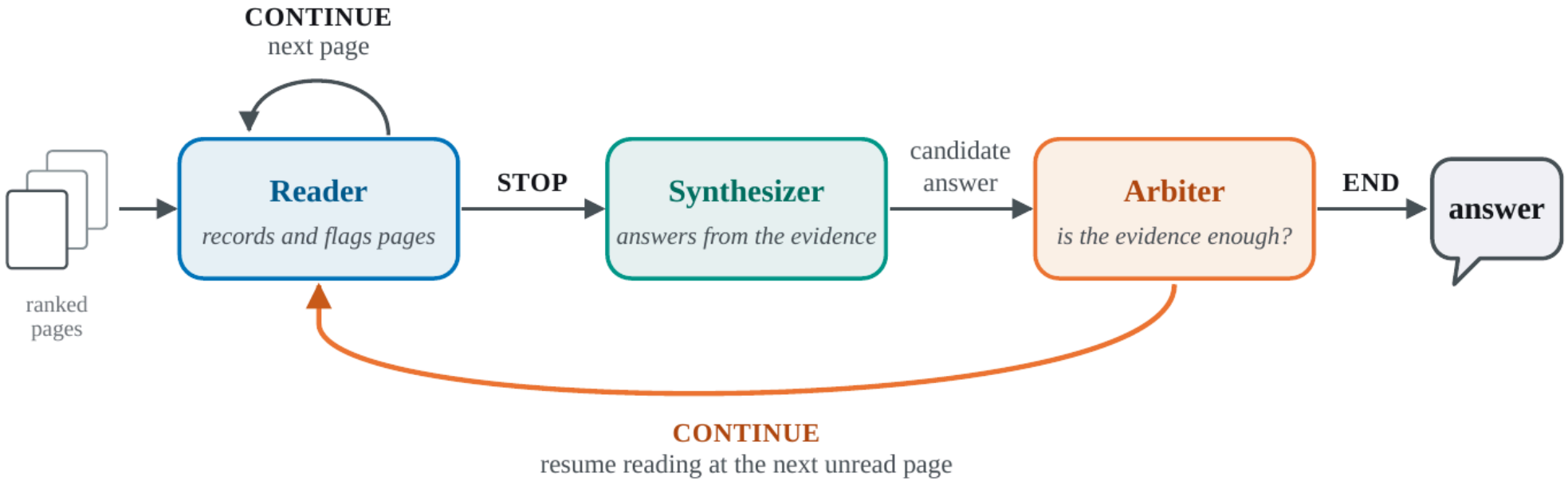
Questions?

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From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

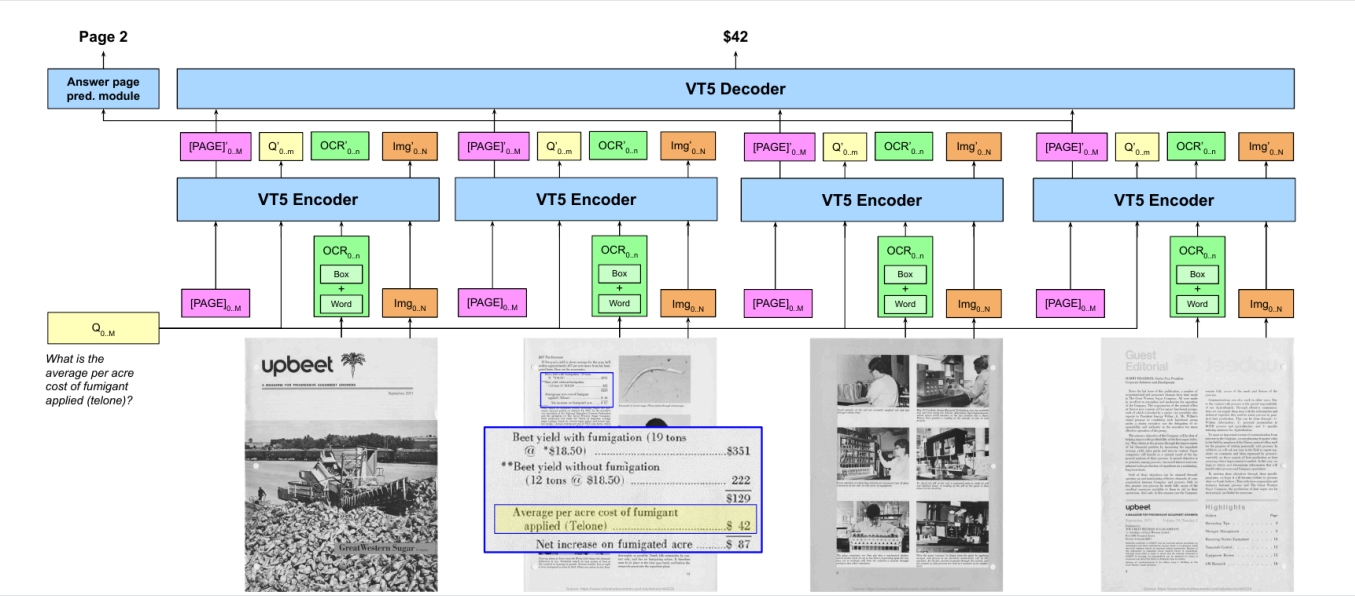
Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	**ACC**	UND	REA	LOC	**ACC**
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9[a]	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8[a]	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8[a]	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8[a]	65.7	53.0	**53.6**	60.0
DocTrace	✓	✓	53.2	**41.1**	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	**60.5**	39.2	**76.9**	**56.5**	**67.2**	**66.3**	47.9	**61.3**

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Page-to-Document Encoding: Hi-VT5

Page-to-document encoding | MLLM-centric adaptation | Retrieval-augmented | Adaptive-trajectory



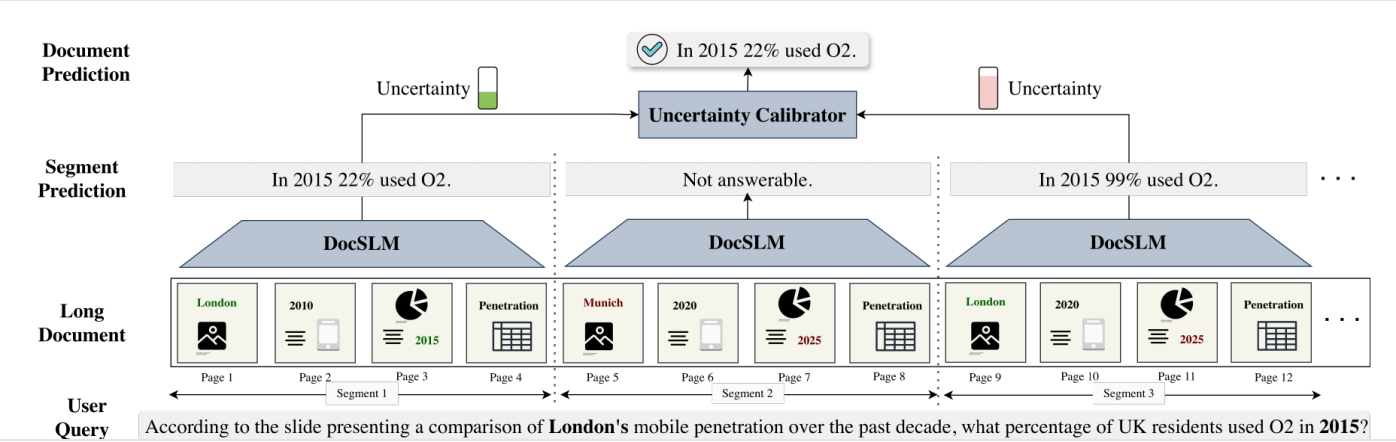
Hi-VT5 architecture. Each page is encoded into [PAGE] tokens, and the decoder reads the tokens of all pages to generate the answer. A second module predicts the answer page. From Tito et al., Pattern Recognition 2023.

HOW IT WORKS

- Builds on **T5**, a text encoder-decoder, and adds visual and layout information.
- Each page is **encoded separately**, with the question, the recognised words and their positions, and patches of the page image.
- Ten learnable **[PAGE] tokens** summarise what on that page is relevant to the question.
- The decoder reads the **joined page tokens**, not the full pages, and generates the answer.

MLLM-Centric Adaptation: DocSLM

Page-to-document encoding | **MLLM-centric adaptation** | Retrieval-augmented | Adaptive-trajectory



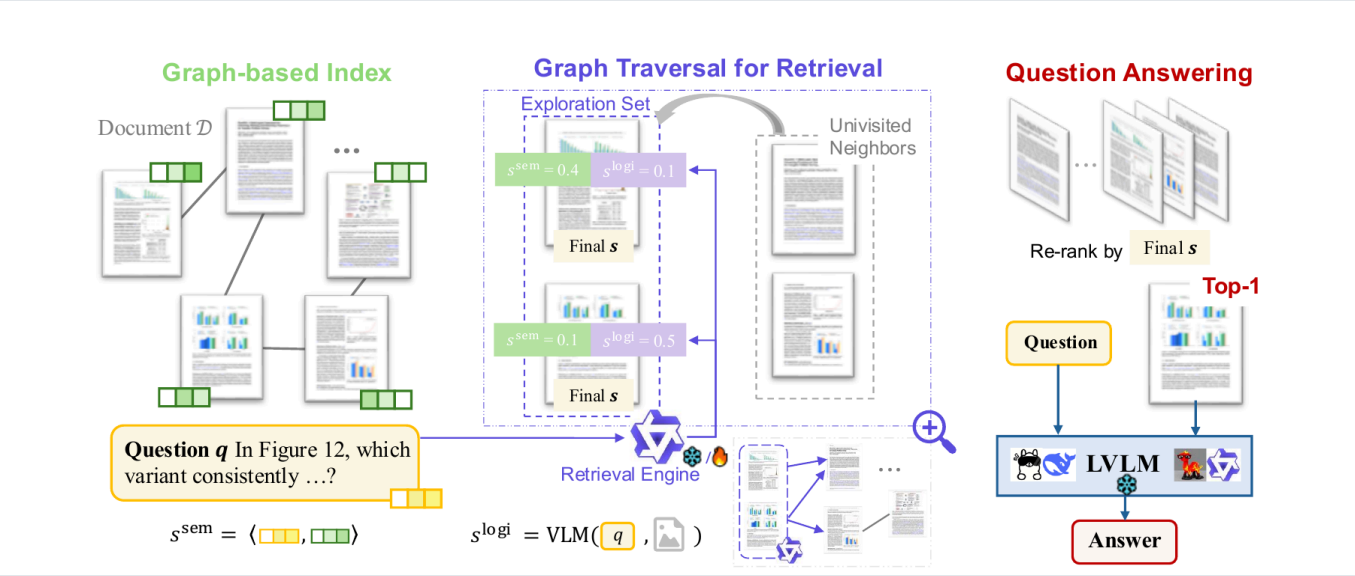
Streaming abstention in DocSLM. The document is split into segments. Each segment gives an answer or abstains, and an uncertainty calibrator keeps the answer with the lowest uncertainty. From Hannan et al., CVPR 2026 Findings.

HOW IT WORKS

- A **small vision-language model** for long documents **under limited memory**.
- **Hierarchical multimodal compressor.** Each page's visual, text and layout information is jointly encoded into a **fixed 576 tokens**.
- **Streaming abstention.** The document is split into segments and processed one at a time. Each segment answers or abstains.
- An **uncertainty calibrator** keeps the answer with the lowest uncertainty.

Retrieval-Augmented: MoLoRAG

Page-to-document encoding | MLLM-centric adaptation | **Retrieval-augmented** | Adaptive-trajectory



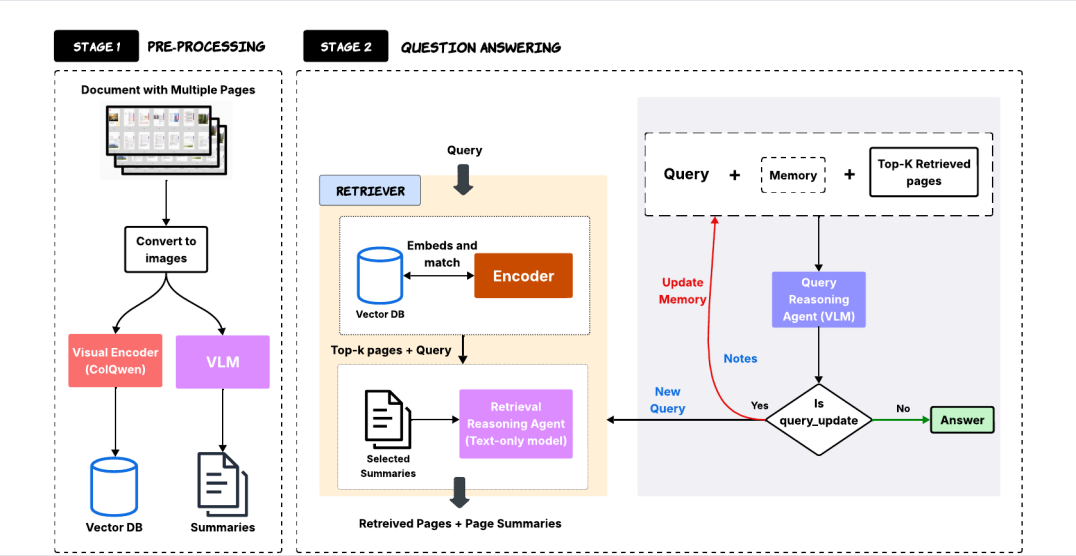
The MoLoRAG framework: a graph-based index over pages, graph traversal for retrieval using semantic and logical relevance scores, then question answering over the top pages. From Wu et al., EMNLP 2025.

HOW IT WORKS

- **Graph-based index.** A visual retriever represents each page, and pages are connected by their similarity.
- **Graph traversal.** The search starts from pages similar to the question. A small vision-language model helps score relevance, and the top K pages are kept.
- **Question answering.** The top K pages go to a separate vision-language model, the reasoner.

Adaptive-Trajectory: SimpleDoc

Page-to-document encoding | MLLM-centric adaptation | Retrieval-augmented | **Adaptive-trajectory**



SimpleDoc. Stage 1 indexes each page as an embedding and a summary. Stage 2 retrieves and re-ranks pages, and a reasoning agent answers or updates its memory and query. From Jain et al., EMNLP 2025.

HOW IT WORKS

- **Pre-processing.** Every page is indexed twice: as a **visual embedding**, and as a short **summary** written by a vision-language model.
- **Dual-cue retrieval.** Pages are shortlisted by embedding similarity, then a language model **re-ranks them using the summaries**.
- **Reasoning agent.** It reads the retrieved pages with a **working memory** of earlier notes, then either answers or **issues a refined query** to retrieve more.
- The loop ends when it answers, decides the question is not answerable, or reaches a step limit.

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Datasets and Evaluation

PRIMARY

MMLongBench-Doc

135	48.3	847	244
documents	mean pages	answerable	unanswerable

- Annotated with **evidence pages, evidence sources and answerability**, which the experiments need.
- 367 of the answerable questions need two or more pages.

COMPLEMENTARY

LongDocURL

396	85.6	2,325
documents	mean pages	questions

- **Born-digital only**, so answers are often in the text layer, even for figure questions.
- Used for the modality-ceiling result only: its evidence pages are not a minimal set.

Judge: Gemini-2.5-Flash at temperature zero returns correct, incorrect or abstained. Agreement is **96.0%** with GPT-4o and **98.3%** with a human annotator.

Modality Ceiling

	Abstention				Accuracy			
Evidence	**T**	**TL**	**TLV**	**V**	**T**	**TL**	**TLV**	**V**
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	36.9	44.4	**51.0**	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	**49.2**	43.2
Table	15.9	7.2	6.7	**9.1**	35.6	**44.7**	**44.7**	34.1
Chart	32.6	**28.1**	7.3	6.2	16.3	16.9	**42.7**	40.4
Figure	**36.6**	24.8	**7.6**	7.9	14.5	22.1	**42.4**	39.0
All	26.3	17.2	6.2	7.1	29.0	35.7	**49.5**	42.6
LongDocURL								
Text	4.6	3.5	**1.5**	**1.6**	71.8	75.6	**78.2**	70.0
Layout	2.1	1.6	0.7	0.9	45.0	49.2	**51.3**	49.3
Table	5.0	3.2	1.1	0.8	57.7	**65.6**	63.6	52.4
Figure	**6.7**	**8.6**	**1.5**	1.3	59.9	62.8	**69.9**	61.9
All	4.7	4.3	1.3	1.3	60.3	64.5	**67.3**	59.4

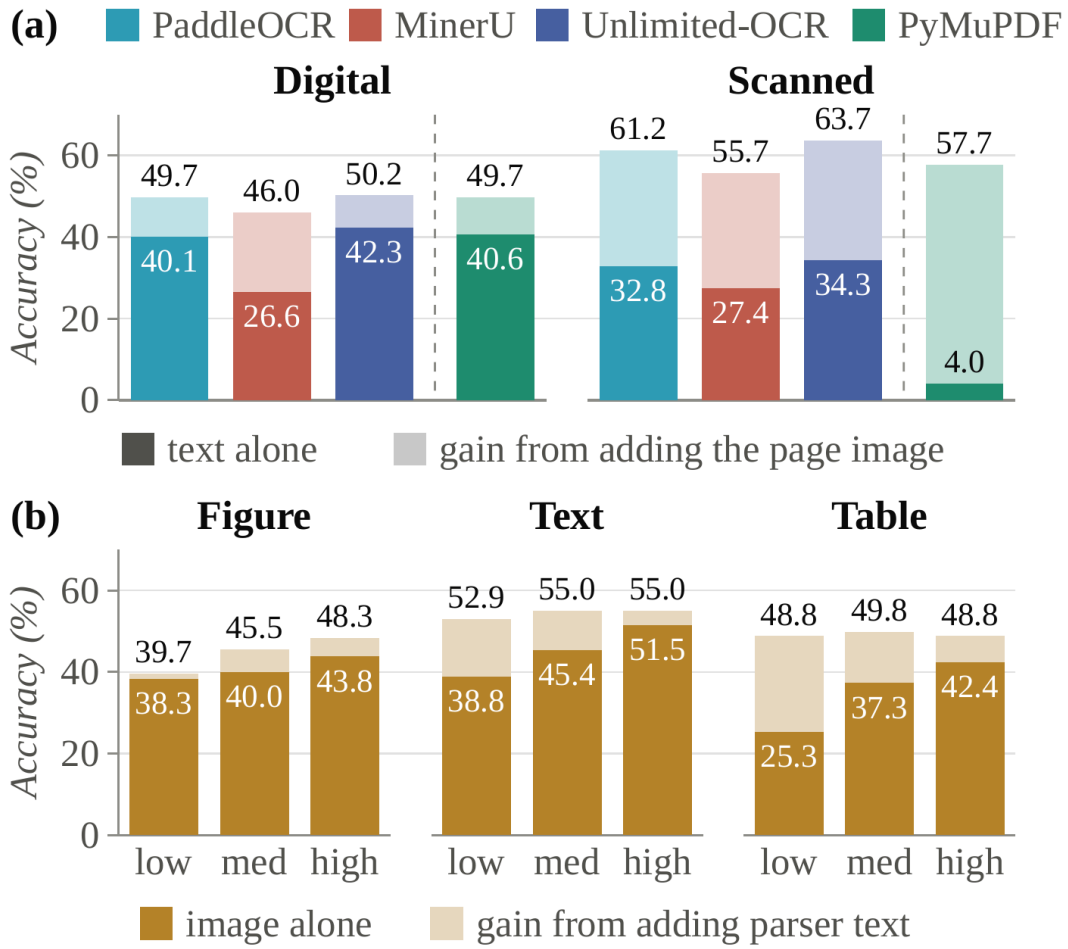


REPRESENTATION

Evidence accessibility depends on modality

- On MMLongBench-Doc, text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**.
- Images alone stay below text plus images: 42.6% against 49.5%.
- On born-digital LongDocURL, adding vision to parser text gives only **2.8 points** overall, 7.1 on figure questions.

Conversion Fidelity



Encode
parser, resolution

Retrieve
gold pages

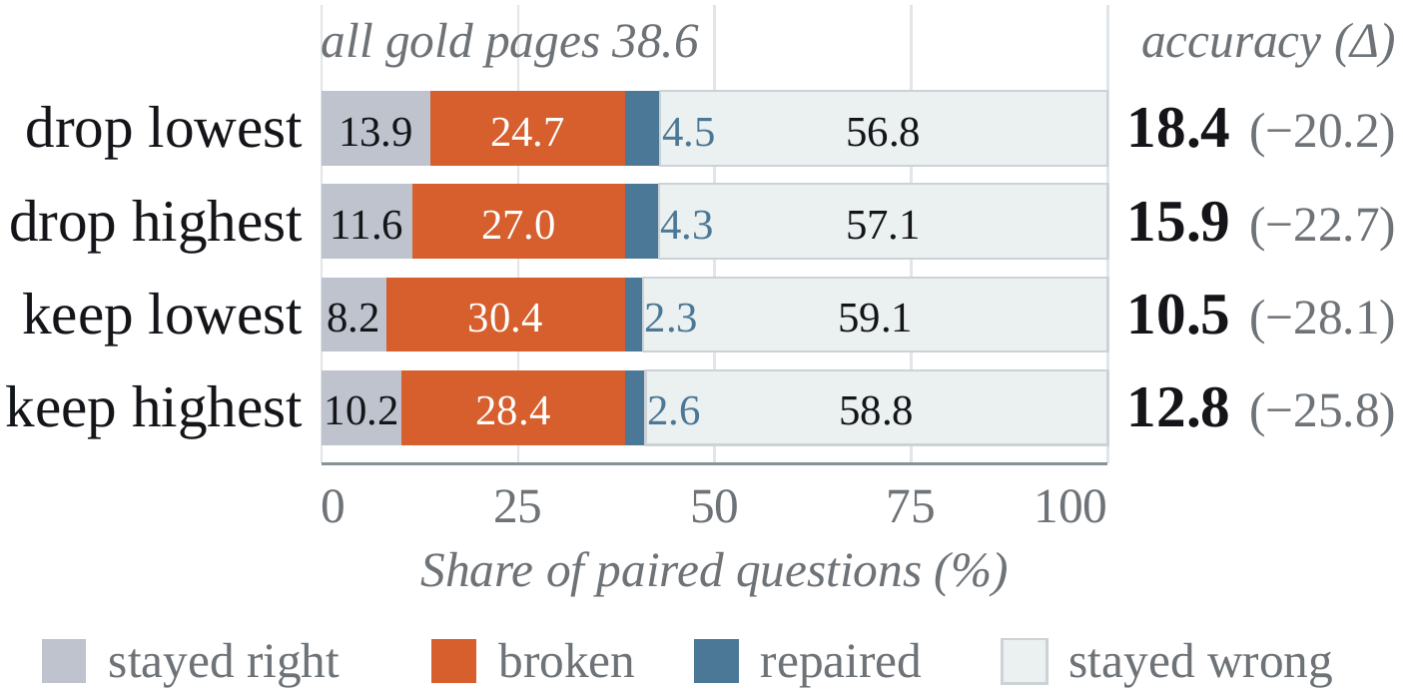
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- Adding page images cuts the spread across parsers from **15.7 to 4.2 points** on digital documents.
- Higher resolution helps images alone far more than text plus images.
- Scanned documents stay sensitive to the parser, and figure questions to resolution.

Evidence Coverage



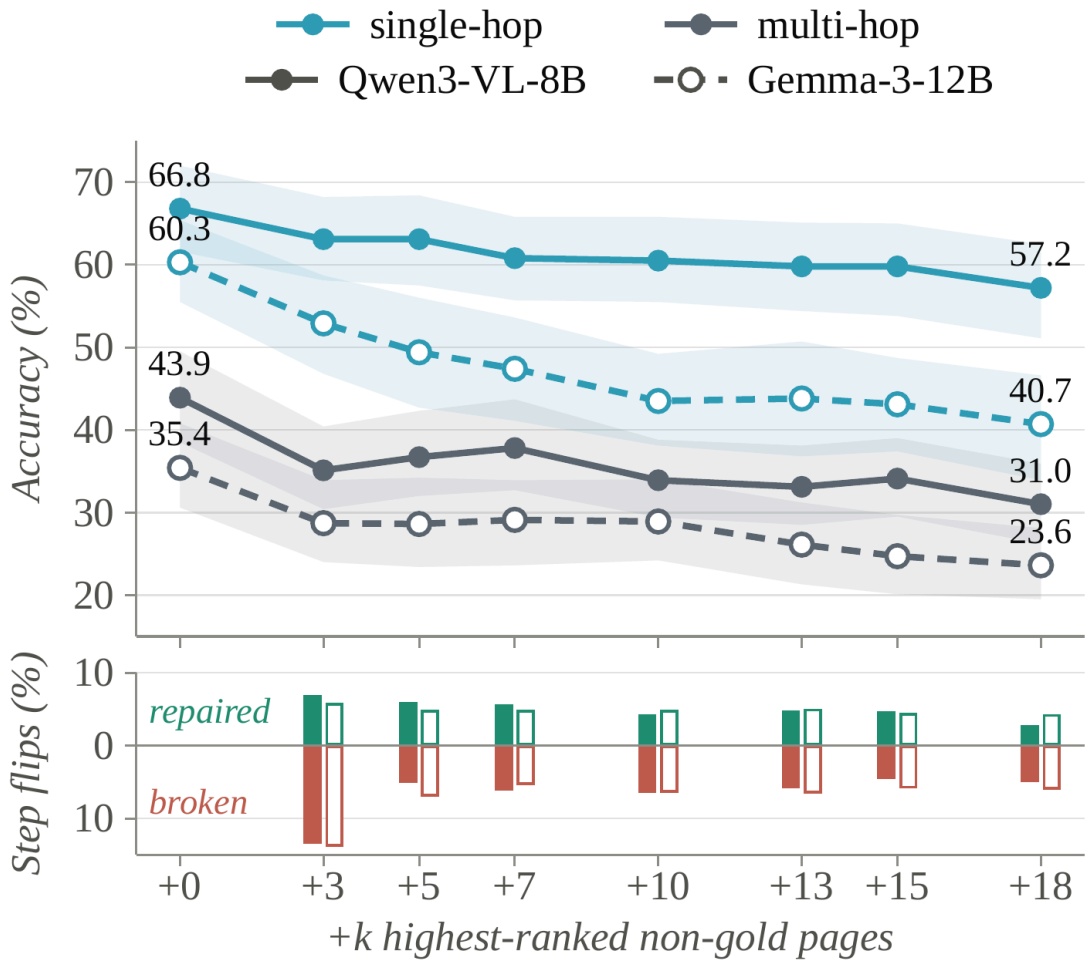


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** (lowest-ranked) or 15.9% (highest-ranked).
- Correct answers break far more often than wrong ones are repaired.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other pages

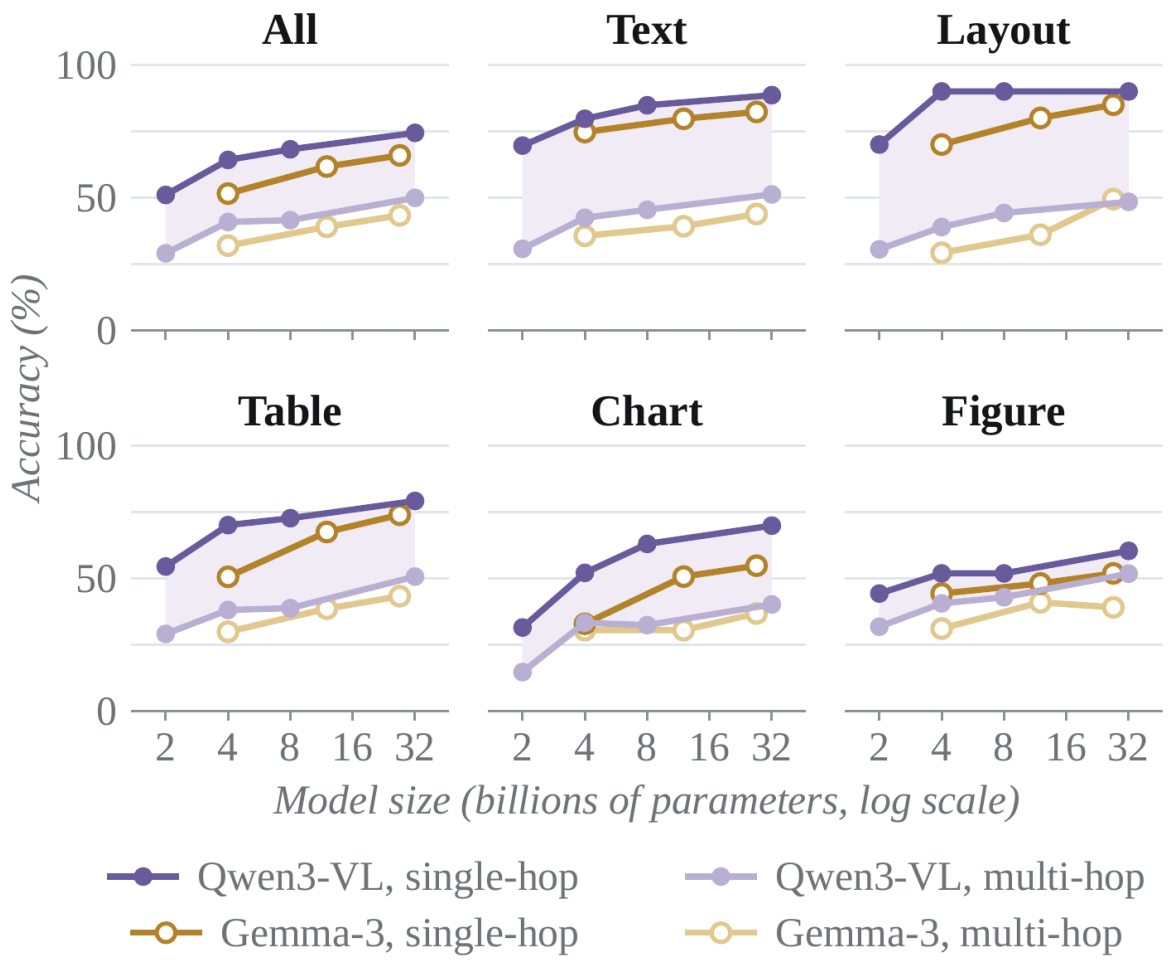
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Qwen3-VL-8B: multi-hop falls from **43.9% to 35.1%** after three extra pages. Single-hop declines gradually, 66.8% to 57.2%.
- Gemma-3-12B shows the same trend: multi-hop 35.4% to 23.6%, single-hop 60.3% to 40.7%.
- Every gold page is still there. Extra context both breaks and repairs answers.

Evidence Integration



Encode
fixed

Retrieve
gold pages

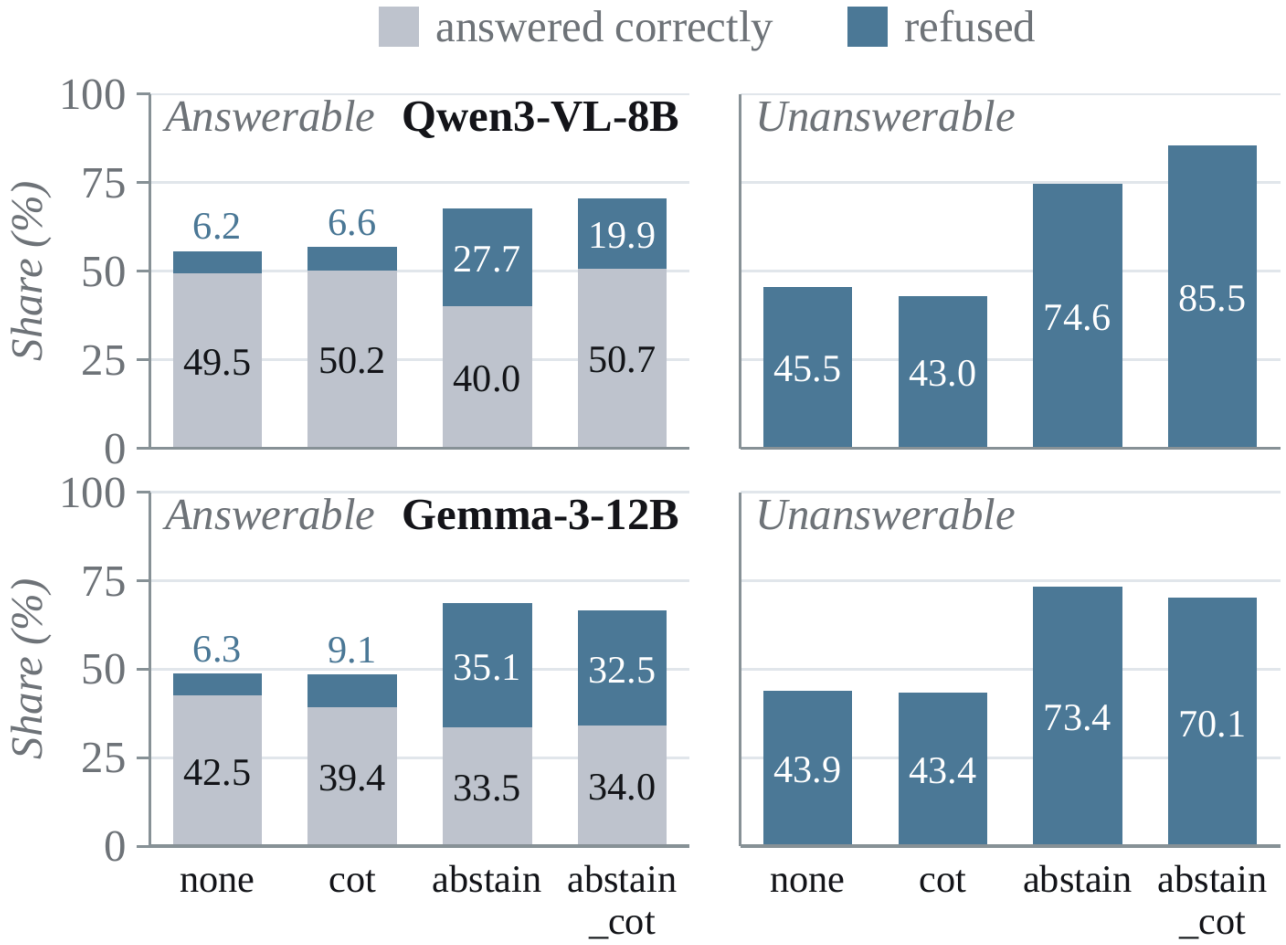
Answer
model size

REASONING

Scaling helps, but the multi-hop gap persists

- Qwen3-VL from 2B to 32B: multi-hop rises from **29.0% to 49.9%**, single-hop from 51.0% to 74.4%.
- The gap between them stays roughly the same.
- Gemma-3 shows the same separation.

Response Calibration



Encode
fixed

Retrieve
gold pages

Answer
prompt mode

REASONING

Reasoning before abstaining limits false refusal

- Qwen3-VL-8B, abstain: unanswerable refusal **45.5% to 74.6%**, but false refusal **6.2% to 27.7%** and accuracy 49.5% to 40.0%.
- Adding CoT: false refusal 19.9%, accuracy 50.7%, unanswerable refusal 85.5%.
- Gemma-3-12B gains much less from CoT: false refusal 35.1% to 32.5%.

Representation: Design and Deployment

Is the evidence in the input?

Modality ceiling

two inline icons, parsed as <img>

Q What are the colors of the icons that the users touch to move the app up and remove the app respectively?

Gold gray, red

T	✗	"The document does not specify the colors"
TL	✗	"it does not mention any visual details"

▼ **add the page image**

V	✓	"The icon is gray. … The icon is red."
TLV	✓	"gray (a gray upward arrow) … red"

Colour exists only in pixels; a text input leaves the reasoner nothing to read.

Conversion fidelity

scanned slide, text by parser

Q What is the name of the company that created this presentation?

Gold AVALAUNCH MEDIA

PaddleOCR	✗	"Asterks" (parse: KICK ASTERKS)
MinerU	✗	"does not provide any evidence"

▼ **re-convert, or add the image**

Unlimited	✓	"AVALAUNCH MEDIA"
with image	✓	"Avalaunch Media … in the banner"

The error sits in the transcript, not the reasoner, and each parser fails differently.

DESIGN — **Complementary modalities** recover what one input loses: prioritise reliable embedded text, add images when appearance matters or extraction is unreliable, re-parse when the text is inadequate.

DEPLOYMENT — Embedded text plus images matches parser text plus images (**50.1% against 49.5%**) with fewer tokens. Images cost **26.3 s** of prefill against 1.8 s, so set modality and selection depth together.

Selection: Design and Deployment

Is the right evidence in context?

Evidence coverage

author list on pages 13, 14

Q How many chapter authors of this report come from Columbia University?

Gold 3

page 13	✗	“two” (Horton, Kinney)
page 14	✗	“one” (Garcia-Pando)

▼ **add the complementary page**

both pages	✓	“three” (Horton, Kinney, Garcia-Pando)

Each page is relevant, yet neither suffices: the answer is a property of the page set.

Distractor exposure

10 gold pages of a 20-page deck

Q How many lateral bar graphs are present in the document?

Gold 10

+3 pages	✗	“there are 12 lateral bar graphs”
+5 pages	✗	“there are 13 lateral bar graphs”

▼ **remove competing pages, or read on**

gold pages	✓	“there are 10 lateral bar graphs”
+7 to +18	✓	“there are 10 lateral bar graphs”

A few added pages inflate the count; the gold set, or more context, counts right.

DESIGN — Prioritise **collective evidence coverage** while controlling distractors: structured navigation, summary-based reranking, or iterative retrieval.

DEPLOYMENT — ColQwen3 lifts recall at 5 from **58.1% to 81.0%** over BM25, but needs a GPU and 110 ms per query against 0.3 ms. Judge retrieval depth by **end-to-end answers and latency**, not recall alone.

Reasoning: Design and Deployment

Is the evidence used correctly?

Evidence integration

a diagram, an ablation table

Q If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting?

Gold 0.5%

2B	✗	"2.4%" (reads another table)
4B	✗	"−9.6%" (right table, wrong row)
8B	✗	"(75.9 − 74.9) / 75.9 … ≈ 1.32%"
▼ a larger reasoner		
32B	✓	"75.9 − 75.4 = 0.5"

Each size breaks a different step of one chain: locate the table, read the cell, compute.

Response calibration

two tables, two pages

Q How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

Gold 16.7%

abstain	✗	"Not answerable" (false refusal)
▼ reason before deciding		
abstain_cot	✓	"79.9% − 63.2% = 16.7%"
none	✓	"… 16.7 percentage points lower"
cot	✓	"Step 1: … Answer: 16.7%"

The question is answerable; the abstain instruction turns multi-step difficulty into refusal.

DESIGN — **Match capacity and inference strategy to the operation.** Separating answer construction from verification helps, at the cost of extra computation.

DEPLOYMENT — Quantisation keeps accuracy at a **smaller footprint but is not faster**: under a memory limit prefer a quantised larger model, under a latency limit fewer reasoning steps or a faster backbone.

Deployment: Accuracy and Cost

Representation	Acc.	Text	Vis.	Tokens	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	50.1	500	459	959	–	–
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1
Retriever	**R@1**	**R@5**	**R@10**	**Idx. (s)**	**Qry. (ms)**	**GB**
BM25	32.7	58.1	68.1	0.02	0.3	CPU
BGE-M3	32.8	61.3	72.4	0.01	47	1.2
Qwen3-Emb.	33.5	63.9	74.8	0.19	89	13.1
ColModernVBERT	47.1	74.0	82.7	0.91	41	5.1
ColQwen2.5	49.2	77.2	84.8	1.50	293	8.6
ColQwen3	53.7	81.0	87.9	0.57	110	9.4
Reasoner	**T**	**TL**	**TLV**	**V**	**GB**	**Time (s)**
Qwen3-VL-8B, bf16	31.9	38.4	52.4	45.5	17.5	30.7
8-bit	32.5	37.9	53.6	45.7	10.3~	43.5
4-bit	31.8	39.4	51.8	45.5	6.8~	35.4
Qwen3-VL-32B, 4-bit	37.0	42.1	59.1	52.3	20.0~	47.5
InternVL3-8B	19.3	25.3	32.6	24.1	15.9	5.3
MiniCPM-V-4.5	30.7	38.1	60.1	53.3	17.4	47.2
GLM-4.6V-Flash	25.9	32.9	42.7	30.9	20.6	19.8
Llama-3.2-11B-V	26.5	34.3	33.3	25.3	21.3	10.4
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4	8.0

PRACTICAL DISCUSSION

Every choice has a cost side

- **Representation:** embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost about 26 s of prefill against 1.8 s.
- **Retriever:** ColQwen3 reaches 81.0% recall at 5 against 58.1% for BM25, but needs a GPU and 110 ms per query against 0.3 ms.
- **Reasoner:** 4-bit keeps accuracy at a smaller footprint (51.8% in about 6.8 GB) but is not faster.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

Check whether the answer is supported

Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

		Tokens			Latency (s)		Peak
	Turns	Input text	Input visual	Output	Prefill	Decode	VRAM (GiB)
`reader`	11.5	5,030	816	697	1.8	46.8	7.2
`synthesizer`	1.2	9,307	2,018	463	3.8	34.7	8.1
`arbiter`	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.